

YU FAN

Male | Age 44 | +86 13918677360 | ✉ yuff100@163.com 19 Years of Experience | Technical Director | Preferred
Location: Shanghai

Summary

- Open to work location; business travel acceptable
- 20 years of high-availability, high-performance backend R&D
- 10+ years managing teams of 30+ engineers
- 4 years of overseas work experience
- Tech stack: C / C++ / Go / Python
- Server-side architect, DevOps architect, innovation prototyping
- Carrier core network, 4G/5G RAN, NPN private network solutions
- VPP/DPDK high-performance media server architecture
- Cloud-native architecture, CI/CD
- Entrepreneurial mindset for business growth and team building
- VIA Strengths: Teamwork, Multi-perspective, Fairness, Humility, Excellence
- MBTI: ENFJ (Extraversion, Intuition, Feeling, Judging)
- Personal website: www.deepnomind.com
- InfoQ Featured Creator – 2021 & 2022

Work

Mavenir Systems

May 2008 - Present

R&D Group Director

Promotion: Software Engineer (2008) → Senior Software Engineer (2009) → Lead Software Engineer (2011) → R&D Manager (2014) → R&D Director (2017) → R&D Group Director (2020)

- Team building, performance management, project planning, and architecture design
- R&D of IMS core network elements: P-CSCF, IBCF, ATCF, E-CSCF, I-CSCF, S-CSCF, TAS (Linux / C / C++)
- Media gateway middleware for WebRTC, SBC, VoLTE/VoWiFi (Linux / C / C++ / DPDK / VPP)
- Layer-7 load balancing architecture supporting NFV evolution
- 4G/5G RAN CU design and implementation; led cloud-native architecture evolution (Linux / C / Go)
- Built CI/CD team, defined processes, automated test and deployment scripts (Python / Jenkins)
- Built 4G/5G RAN architect team; managed NPN private network automated deployment
- Innovation prototyping: RAN DU uBPF integration, AI real-time voice translation
- Completed 5G CUUP microservice transformation; separated 4G CU control plane and user plane
- Built RU PnP OAM microservice supporting 8,000+ RUs dynamic onboarding
- Introduced VPP/DPDK for user-plane acceleration (SBC, 4G/5G CUUP)
- Planned 10+ CI/CD pipelines for multiple R&D and QA teams
- Built proprietary automated testing framework for 4G CU, 5G CUUP, SDaaS, IMS, SBC
- Built AI real-time voice translation server on UAG platform (cloud + on-device inference; deployed to T-Mobile US)

Motorola

May 2007 - May 2008

Software Engineer

- Developed iDEN handset applications
- Developed automated test scripts based on STRIDE
- Developed PoC (Push-to-talk over Cellular) application server
- Designed and implemented PTT server
- Designed test cases for multiple handset features

Projects

AI Voice Application Service

Jul 2025 - Present

Architect

- Designed real-time voice translation service architecture integrated with carrier IMS core network
- Led 5-person team through technology selection, prototype design, and architecture delivery
- Optimized voice data transmission with BPF technology; end-to-end latency under 200ms
- Dual-mode architecture: cloud inference + on-device inference
- 99.99% system availability supporting millions of concurrent users

Radio Access Network

Oct 2019 - Jun 2025

Group Director

- Led 4G/5G base station R&D with cloud-native architecture implementation
- Led infrastructure setup and business requirement translation to technical solutions
- Monitored project progress ensuring on-time delivery
- Regularly assessed and optimized system architecture

Private Wireless Network Solution (NPN)

Apr 2022 - Dec 2022

R&D Director

- Led NPN project R&D; drove automated deployment solutions
- Integrated O-RAN/5GC with IMS core network
- Managed Web GUI development and monitoring system design
- Achieved minute-level IMS core network deployment in private networks

Unified Access Gateway (UAG)

Jan 2013 - Oct 2019

R&D Manager

- Led 12-person team in media server R&D
- Developed RTP, MSRP, IPSec, H.248 protocols
- Optimized media server performance with DPDK/VPP
- Customers: T-Mobile, DTAG, Reliance, Tele2, AMX and other major global carriers

IMS Core Network

May 2008 - Jan 2013

Technical Lead

- R&D for I-CSCF, S-CSCF, IP-SM-GW, RCS, SMSC, MMSC
- Designed and developed SIP, CDR, Call Control modules
- Addressed signaling storms and data transmission latency under high concurrency
- Customers: T-Mobile, DTAG, Vietnam Telecom; serving hundreds of millions of users

Motorola Handsets

Apr 2007 - May 2008

Software Engineer

- Developed iDEN handset voice communication applications
- Built automated testing framework based on STRIDE
- Optimized PoC cluster voice transmission protocol and scheduling algorithms

Skills

C / C++

10+ years in telecom system development, high-performance backend and core networks

Go / Python / Java

Cloud-native services and toolchain development

Linux

Kernel performance tuning and BPF technology

DPDK / VPP

High-performance data plane acceleration

IMS Core / VoLTE / VoWiFi / WebRTC / SBC

Full-stack core network elements and voice/data transport

CI/CD / Jenkins / Microservices / Cloud-Native

DevOps and engineering efficiency

Education

Nanjing University of Posts and Telecommunications

2004 - 2007*Master's Degree*

Computer Application Technology

Nanjing Normal University

2000 - 2004*Bachelor's Degree*

Electronic Information Engineering

Certificates

- College English Test Band 6 (CET-6)
- College English Test Band 4 (CET-4)
- National Computer Level 3 Certificate
- Google Certified: Project Management Certificate
- [DeepLearning.AI](#): AI For Everyone
- Google Certified: AI Certificate
- IBM Certified: AI Certificate
- Coursera Certified: Soft Skills, Communication Skills, Time & Stress Management, Conflict Management

Honors

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